Lesson 10: Expected Values

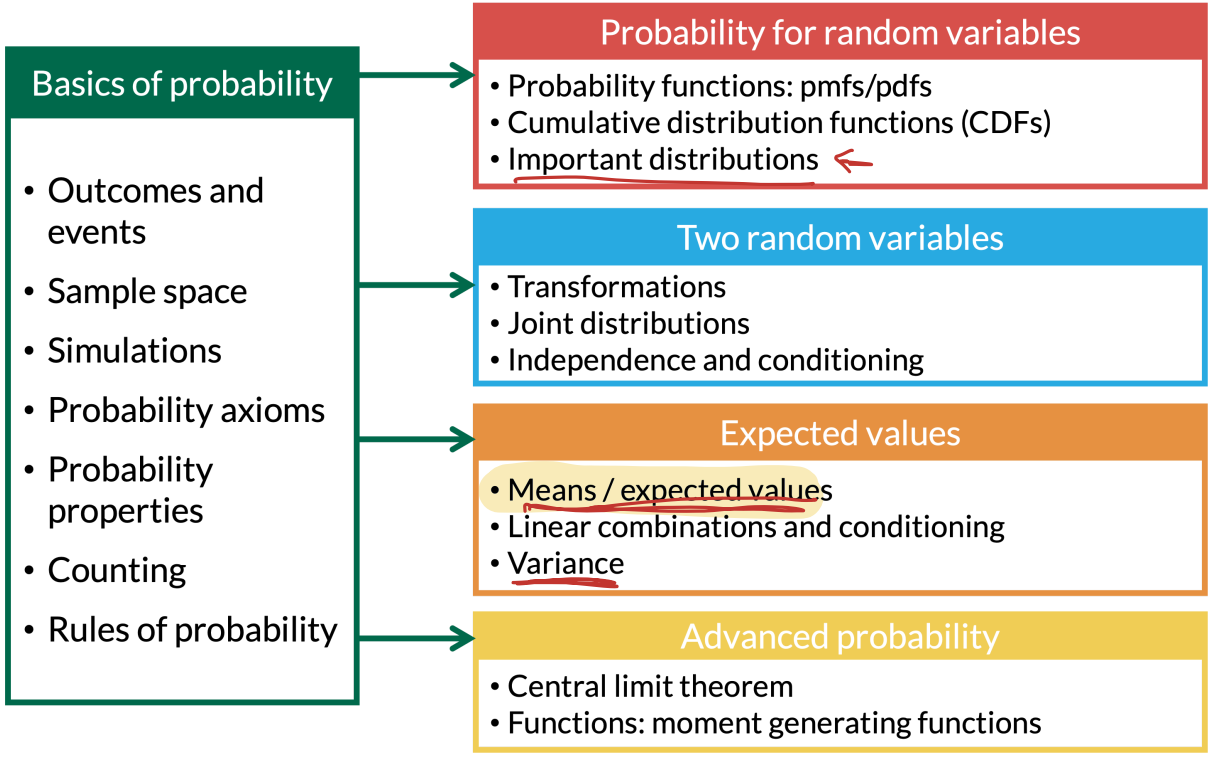
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Learning Objectives

1. Define the expected value for discrete and continuous RVs
2. Calculate the expected value (mean) of a single discrete RV
3. Calculate the expected value (mean) of a single continuous RV
4. Calculate expected value for the sum of discrete or continuous RVs

Where are we?


Basics of probability
• Outcomes and events
• Sample space
• Simulations
• Probability axioms
• Probability properties
• Counting
• Rules of probability
Probability for random variables
• Probability functions: pmfs/pdfs
• Cumulative distribution functions (CDFs)
• Important distributions
Two random variables
• Transformations
• Joint distributions
• Independence and conditioning
Expected values
• Means / expected values
• Linear combinations and conditioning
• Variance
Advanced probability
• Central limit theorem
• Functions: moment generating functions

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What is an expected value?

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

Definition: Expected value of discrete RV

The expected value of a discrete RV X that takes on values $x_1, x_2, \ldots, x_n$ is

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i p_X(x_i)$$

weight

where n can be ∞

Definition: Expected value of continuous RV

The expected value of a continuous RV X is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx$$

where we adjust the integrand based on the bounds of X

weighted avg

expected values are not necessarily a possible outcome

Our good and fair friend, the 6-sided die

Example 1

Suppose you roll a fair 6-sided die. What value do you expect to get?

Let X be the value rolled. Since the die is fair, $p_X(x) = \frac{1}{6}$ for $x = 1, 2, \ldots, 6$.

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{i=1}^{6} x_i p_X(x_i) \\
&= 1\left(\tfrac{1}{6}\right) + 2\left(\tfrac{1}{6}\right) + 3\left(\tfrac{1}{6}\right) + 4\left(\tfrac{1}{6}\right) + 5\left(\tfrac{1}{6}\right) + 6\left(\tfrac{1}{6}\right) \\
&= \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5
\end{aligned}$$

- 3.5 is the long-run average: if we roll the die many, many times, the average of the rolls will be close to 3.5

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Our good and not-so-fair friend, the 6-sided die

Example 2

Suppose the die is 6-sided, but not fair. And the probabilities of each side are distributed as:

x	$p_X(x)$
1	0.10
2	0.05
3	0.02
4	0.30
5	0.50
6	0.03

What value do you expect to get on a roll?

$$E(X) = \sum_{i=1}^{6} x_i \, p_X(x_i)$$

$$= 1(0.1) + 2(0.05) + 3(0.02) + 4(0.3) + 5(0.5) + 6(0.03)$$

$$= 4.14$$

Let's simulate our not-so-fair die

- Same idea as Lesson 2: simulate many rolls, then summarize
- `prob` gives the probability of each side, like the unfair coin in Lesson 1

```
1  set.seed(1026)
2  reps <- 10000
3  die_probs <- c(0.10, 0.05, 0.02, 0.30, 0.50, 0.03)
4
5  X_sims <- tibble(sim = 1:reps) |>
6    mutate(X = sample(1:6, n(), replace = TRUE,
7                      prob = die_probs))
```

- Average of the 10,000 simulated rolls:

```
1  X_sims |>
2    summarise(average = mean(X))
```

```
# A tibble: 1 × 1
  average
    <dbl>
1    4.16
```

From Example 2, $\mathbb{E}[X] = 1(0.10) + 2(0.05) + 3(0.02) + 4(0.30) + 5(0.50) + 6(0.03) = 4.14$

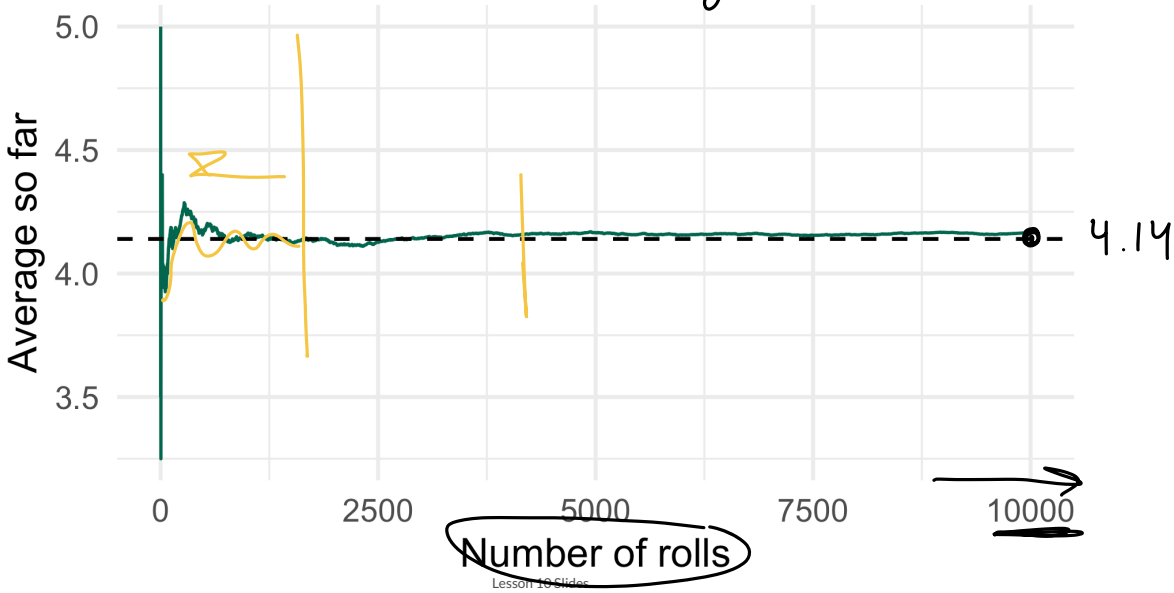
The simulated average is very close!

The running average settles at $\mathbb{E}[X]$

- The **expected value** is the long-run average: after many rolls, the average of the rolls gets close to 4.14 (dashed line)

▶ Show/Hide Code for the running average plot

reps → ∞
sim avg → exact E(X)



Expected value of a Bernoulli distribution

Example 3

Suppose

$$X = \begin{cases} 1 & \text{with probability } p \quad \text{(success)} \\ 0 & \text{with probability } 1-p \quad \text{(failure)} \end{cases}$$

Find the expected value of X.

$$E(X) = \sum_{i=1}^{2} x_i P(X = x_i) = 1(p) + 0(1-p)$$

$$= p$$

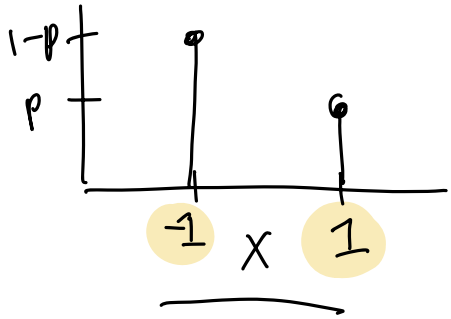
Let's slightly change our random variable

Example 4

Suppose

$$X = \begin{cases} 1 & \text{with probability } p \\ -1 & \text{with probability } 1-p \end{cases}$$

Find the expected value of X.



#s attached to success & failure

$$E(X) = \sum_{i=1}^{2} x_i \, p(X = x_i) = 1(p) + (-1)(1-p)$$

$$= p - 1 + p = 2p - 1$$

$p = \frac{1}{2}$: $E(X) = 0$

$p > \frac{1}{2}$: $E(X) > 0$

$p < \frac{1}{2}$: $E(X) < 0$

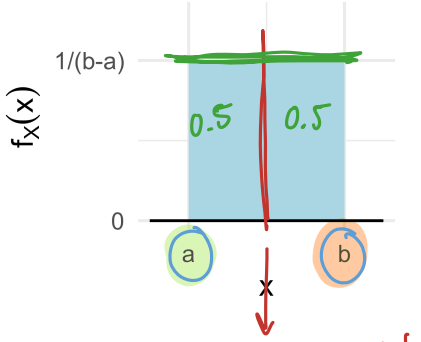
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Expected Value of the Continuous Uniform Distribution

Example 5

Let $f_X(x) = \frac{1}{b-a}$ for $a \leq x \leq b$. Find $\mathbb{E}[X]$.



$E(X) = \frac{a+b}{2}$

$$E(X) = \int_a^b x\, f_X(x)\, dx$$

$$= \int_a^b x \left(\frac{1}{b-a}\right) dx$$

$$= \left(\frac{1}{b-a}\right)\frac{1}{2}x^2 \Big|_{x=a}^{x=b}$$

$$= \left(\frac{1}{b-a}\right)\frac{1}{2}\left(b^2 - a^2\right)$$

$x^2 - y^2 = (x+y)(x-y)$

$$= \frac{(b+a)(b-a)}{2(b-a)}$$

$$= \frac{b+a}{2}$$

Expected Value of the Exponential Distribution

Example 6

Let $f_X(x) = \lambda e^{-\lambda x}$, for $x > 0$ and $\lambda > 0$. Find $\mathbb{E}[X]$.

Integrating by Parts

$$\int_a^b u\,dv = uv\Big|_a^b - \int_a^b v\,du$$

$u = \lambda x$ $\quad dv = e^{-\lambda x}\,dx$

$du = \lambda\,dx$ $\quad v = \frac{-1}{\lambda} e^{-\lambda x}$

$\int 1\,dv = \int e^{-\lambda x}\,dx$

$v = \frac{-1}{\lambda} e^{-\lambda x}$

$\frac{d}{dx}\left(-\frac{1}{\lambda} e^{-\lambda x}\right) = -\frac{1}{\lambda}\cdot(-\lambda) e^{-\lambda x}$

$$E(X) = \int_0^\infty x\, f_X(x)\,dx$$

$$= \int_0^\infty x \lambda e^{-\lambda x}\,dx$$

$$= \lambda x\left(\frac{-1}{\lambda} e^{-\lambda x}\right)\Big|_0^\infty + \int_0^\infty \frac{1}{\lambda} e^{-\lambda x} \lambda\,dx$$

$\lim_{x\to\infty} x e^{-x} = 0$

$$= -x e^{-\lambda x}\Big|_0^\infty + \int_0^\infty e^{-\lambda x}\,dx$$

$$= 0 - 0 + \left(-\frac{1}{\lambda} e^{-\lambda x}\right)\Big|_0^\infty$$

$$= 0 + \left(\frac{1}{\lambda} e^{-\lambda(0)}\right)$$

$$E(X) = \frac{1}{\lambda}$$

Picking up our exponential simulation from Lesson 9

- In Lesson 9 (Example 4), we simulated X with pdf $f_X(x) = 2e^{-2x}$: Example 6 with $\lambda = 2$
- Same code as Lesson 9, so we get the same 10,000 simulated values

exponential w/ $\lambda = 2$

① Lesson 9: found CDF

```
1  set.seed(2026)
2  reps <- 10000
3  exp_sims <- tibble(u = runif(n = reps, min = 0, max = 1)) |>
4    mutate(x = -log(1 - u) / 2)
```

① ② ③

② sample u from uniform cont. dist.

③ finding x from inv CDF of u

inv CDF of $f_X(x)$ w/ $\lambda = 2$

- New step: the average of the 10,000 simulated values

```
1  exp_sims |>
2    summarise(average = mean(x))
```

```
# A tibble: 1 × 1
  average
    <dbl>
1   0.498
```

$\approx \frac{1}{2}$

From Example 6, $\mathbb{E}[X] = \frac{1}{\lambda} = \frac{1}{2}$

The simulated average is very close!

we could also jump straight to rexp() for simulating x

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Revisiting our two card draw

Example 7

Suppose you draw 2 cards from a standard deck of cards *with* replacement. Let X be the number of hearts you draw. Find $\mathbb{E}[X]$.

↳ $P(\heartsuit) = \frac{13}{52} = \frac{1}{4}$

Recall Binomial RV with $n = 2$:

$$p_X(x) = \binom{2}{x} p^x (1-p)^{2-x} \quad \text{for } x = 0, 1, 2$$

Jump to Binomial:

$$E(X) = \sum_{i=1}^{3} x_i P(X = x_i)$$

$$= 0 \cdot P(X=0) + 1\, P(X=1) + 2 P(X=2)$$

$$= 0 \cdot \binom{2}{0} 0.25^0 0.75^2 + 1 \binom{2}{1} 0.25^1 0.75^1 + 2 \binom{2}{2} 0.25^2 0.75^0$$

$Y = \begin{cases} 1 & p \\ 0 & 1-p \end{cases}$ (1: get ♡, 0: no ♡)

★ Binomial (n, p) is the sum of n Bernoulli (p)

↳ $E(Y) = p$

$$X = \sum_{i=1}^{2} Y_i$$

$$E(X) = E\left[\sum_{i=1}^{2} Y_i\right] = E(Y_1 + Y_2) = E(Y_1) + E(Y_2) \quad (2 \cdot \tfrac{1}{4})$$

$$= p + p = 2p = \frac{1}{2}$$

Sums of Random Variables

Theorem: Sum of random variables do NOT need to be independent

For RVs (discrete or continuous) X_i and constants $a_i, i = 1, 2, \dots, n$,

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i].$$

exp val of sum is sum of exp val's

Remark: The theorem holds for infinitely many RVs X_i as well.

- For two RVs, X and Y:
  - We can say $E[X + Y] = E[X] + E[Y]$
  - ... and constant numbers a and b, we can also say $E[aX + bY] = aE[X] + bE[Y]$
  - We can also also say $E[X - Y] = E[X] - E[Y]$, since $b = -1$

Corollaries from Theorem

Function with two constants

For a RV X, and constants a and b,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

$E(aX) + E(b)$

↳ $E(b) = b$

exp val of constant is just the constant

Expected value of sum of identically distributed RVs

If $X_i, i = 1, 2, \ldots, n,$ are identically distributed RV's, then (not nec ind)

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = n\mathbb{E}[X_1].$$

$E(X_1) = E(X_2) = \ldots = E(X_n)$

$\sum E(X_i) = E(X_1) + E(X_2) + \ldots E(X_n)$

$= n\,E(X_1)$

Cost of hotel rooms

Let C_i = cost of room i (tax not inc)

T = total cost of all 30 room

Example 8

A tour group is planning a visit to the city of Minneapolis and needs to book 30 hotel rooms. The average price of a room is $200. In addition, there is a 10% tourism tax for each room. What is the expected cost for the 30 hotel rooms? $E(T)$

$$E(C_i) = \$200$$

$$T = \sum_{i=1}^{30} 1.1\, C_i$$

1.1: 10% tax

$$E(T) = E\left[\sum_{i=1}^{30} 1.1\, C_i\right] = \sum_{i=1}^{30} E(1.1\, C_i) = \sum_{i=1}^{30} 1.1\, E(C_i)$$

$1.1\, C_i$: a X; $1.1\, E(C_i)$: a $E(X)$

exp val of sum is sum of exp val

$$= 1.1 \sum_{i=1}^{30} E(C_i) = 1.1 \sum_{i=1}^{30} \$200 = 1.1(200 + 200 + \ldots + 200)$$

$$= 1.1 \cdot 30 \cdot \$200 = \$6{,}600$$ is expect val of total cost

Let's simulate the cost of hotel rooms (1/2)

- Assume each room is equally likely to cost $180, $200, or $220, so the average price is $200

```
reps <- 10000
n_rooms <- 30
room_prices <- c(180, 200, 220)

rooms <- expand_grid(sim = 1:reps, room = 1:n_rooms) |>
  mutate(price = sample(room_prices, n(), replace = TRUE),
         cost = 1.1 * price)

cost_sims <- rooms |>
  group_by(sim) |>
  summarise(total_cost = sum(cost))
```

$E(C_i) = \$200$

① ② ③ ④ ⑤ ⑥

including tax

summing across 30 rooms

10,000 total_cost's

```
rooms
```

```
# A tibble: 300,000 × 4
     sim  room price  cost
   <int> <int> <dbl> <dbl>
 1     1     1   200   220
 2     1     2   220   242
 3     1     3   180   198
 4     1     4   220   242
 5     1     5   200   220
 6     1     6   200   220
 7     1     7   180   198
 8     1     8   180   198
 9     1     9   220   242
10     1    10   180   198
# ℹ 299,990 more rows
```

price: w/ tax

Let's simulate the cost of hotel rooms (2/2)

- Average total cost over the 10,000 simulations:

```
1 cost_sims |>
2   summarise(average = mean(total_cost))
```

```
# A tibble: 1 × 1
  average
    <dbl>
1   6599.
```

From Example 8, $\mathbb{E}\left[\sum_{i=1}^{30} 1.1X_i\right] = 30(1.1)(200) = \$6600$

- Any room prices with an average of $200 give the same expected total. Try changing `room_prices`!

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